Diseases of the Pituitary & Adrenal Glands

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

Pituitary adenoma, hyperprolactinemia, hypopituitarism, Cushing's, Addison's, hyperaldosteronism

Lecture roadmap: How the pituitary and adrenal glands fit together → Pituitary adenoma → Hyperprolactinemia → Nonfunctioning adenoma (headache and vision) → Hypopituitarism → Adrenal anatomy and feedback → Cushing syndrome vs. Cushing's disease → Why Cushing's causes diabetes → Addison's disease → Why Addison's darkens the skin → Hyperaldosteronism

How the instructor wants you to think: Know what each hormone **normally does**; then you can explain what happens with **too much or too little**. Pair each pituitary hormone with its **target gland's hormone** (ACTH with cortisol, TSH with free T4), and ask whether the response is **appropriate** for the feedback loop. You don't need the logistics of the hormone tests, but you do need the **relationships between the lab values**.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further or a transcript slip cleaned up; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

How the Glands Fit Together

Gland	Hormone	Normal job
Anterior pituitary (receives signals from the hypothalamus)	**TSH**	Stimulates the thyroid → T3/T4
	ACTH	Stimulates the **adrenal cortex** → **cortisol**
	Growth hormone	Growth in children; muscle mass and exercise capacity in adults
	Prolactin	Milk production during lactation
	FSH and LH	Reproductive hormones

Gland	Hormone	Normal job
Posterior pituitary	**ADH** and **oxytocin**	**Made in the hypothalamus**, stored and released by the posterior pituitary. ADH tells the kidneys to **conserve water**.
Adrenal medulla (inner)	Epinephrine, norepinephrine	Not the focus of this lecture
Adrenal cortex (outer)	**Cortisol**	Keeps **energy available** and supports the response to **physical stress**. Controlled by **ACTH** (feedback loop).
	Aldosterone	Regulates **BP and electrolytes** through the kidneys: **keep sodium (and water), excrete potassium**. Controlled mainly by **RAAS and potassium**, not ACTH.

Instructor emphasis: Cortisol and aldosterone come from the same gland but have totally different jobs and different controls. Cortisol is driven by **ACTH**. Aldosterone is driven by **RAAS and potassium**. This distinction explains Addison's vs. a pituitary cause of adrenal insufficiency.

Pituitary Adenoma

A usually **benign** tumor of the **anterior pituitary** CLARIFIED . The focus is less on the tumor and more on its **effects on the body**. It causes problems in **two ways**, sometimes both: **releasing too much of a hormone** and **pressing on nearby structures**.

Types	• **Functioning adenoma** → hormone **excess**. Examples: a **prolactinoma** (too much prolactin); an **ACTH-producing adenoma** (adrenals make too much cortisol). • **Nonfunctioning adenoma** → no clinically significant hormone excess, but the **growth itself** causes problems.

Manifestations (mass effect)	There isn't much room to expand inside the head: • **Headaches** from pressure on surrounding tissue • **Vision loss**: the **optic chiasm** (where some visual nerve fibers cross) sits **just above** the pituitary. Pressure there → loss of **peripheral vision toward the outer sides of both visual fields** (bitemporal hemianopia) ADDED • **Compression of healthy pituitary tissue** → **reduced** production of other hormones. A person can have **excess of one hormone and deficiencies of others**. **Pituitary apoplexy** (rare): **bleeding or loss of blood supply** in the tumor causes sudden damage → **severe headache**, **abrupt vision changes**, dangerous **hormone deficiencies**.
Diagnosis	**MRI** (size, location, pressure on structures); **visual field testing** (finds subtle loss the person hasn't noticed); **blood tests** for hormone excess or deficiency. Look at the **pituitary hormone alongside its target-gland hormone** (ACTH with cortisol; TSH with free T4). A level can look normal but still be an **inappropriate response**.
Impact on health	Depends on tumor size and which hormones are affected: vision changes interfere with daily activities; hormone imbalances affect energy, reproduction, BP, and more. **Benign does not mean harmless.** Location and hormonal effects make adenomas clinically significant.

Pause and Reflect: How can a nonfunctioning adenoma cause headaches and vision changes?
"Nonfunctioning" means the tumor **doesn't release excess hormone**, but it still grows. The pituitary sits in a **small bony space** at the base of the brain (the **sella turcica**). As the adenoma grows it **stretches or presses on nearby pain-sensitive tissues** → **headache**. If it grows **upward**, it compresses the **optic chiasm** → loss of **peripheral vision** on the outer sides of both visual fields. It can also **compress healthy pituitary tissue** → **less normal hormone production**.

Hyperprolactinemia

An elevated **prolactin** level. High prolactin is **expected during pregnancy and lactation**; this section is about elevations **outside** those settings.

Etiology	**Prolactinoma**; certain **medications**; **hypothyroidism**; disrupted **hypothalamus–pituitary connection**.

Pathophysiology	**Dopamine normally holds prolactin release down.** If a drug **blocks dopamine**, or dopamine **can't reach the pituitary**, that restraint is lost and prolactin rises. Excess prolactin **suppresses GnRH** (from the hypothalamus) → **less FSH and LH** → **lower sex hormones**.
Manifestations	• **Irregular or absent menstrual periods** • **Reduced libido**, **erectile dysfunction**, **infertility** • **Galactorrhea**: milk production outside lactation; it can occur in people **assigned male at birth** • Some have reproductive symptoms **without** milk production
Diagnosis	Measure **prolactin** and look for causes: **pregnancy**, **medications**, **thyroid dysfunction**. **Pituitary imaging** if still unexplained.
Impact on health	Long-term low sex hormones → **bone loss**. Effects reach beyond reproduction and milk production.

Hypopituitarism

The pituitary **doesn't release enough** of one or more hormones. Causes: a **tumor compressing** healthy tissue, or damage from **surgery, radiation, injury, or reduced blood supply**.

Pause and Reflect: What deficiencies and manifestations result? Start with the missing pituitary hormone, then think about its target gland.

Missing hormone	Target-gland effect	Manifestations
TSH	Thyroid gets less stimulation → ↓ thyroid hormone	Hypothyroid picture: **fatigue, cold intolerance, constipation, slow heart rate**
ACTH	Adrenal cortex → ↓ **cortisol**	**Weakness, low BP**, trouble keeping **blood glucose** up (especially when fasting or ill); **poor response to illness or injury**
FSH / LH	↓ sex hormones	Irregular or absent periods, reduced libido, ED, **infertility**

Missing hormone	Target-gland effect	Manifestations
Growth hormone	—	Children: affects **growth**. Adults: **reduced muscle mass**, lower **exercise capacity**
ADH (if the hypothalamus or posterior pituitary is involved)	Kidneys lose the signal to **conserve water**	**Large amounts of dilute urine**, **thirst**, **dehydration**

A person may have **one deficiency or several**. Early complaints are often vague ("tired, weak, unwell"); knowing which hormones are missing explains the pattern.

- **Diagnosis:** identify which hormones are involved and whether target glands respond appropriately. **Stimulation tests** check whether the system can increase hormone production when prompted.
- **Cortisol deficiency is especially important**: it can become **life-threatening during physical stress**.

The Adrenal Glands

- Located on top of the kidneys. **Outer cortex**: cortisol, aldosterone, adrenal androgens. **Inner medulla**: epinephrine, norepinephrine. The disorders below involve the **cortex**.
- **Cortisol feedback loop:** hypothalamus → pituitary releases **ACTH** → adrenal cortex makes **cortisol** → rising cortisol **reduces further stimulation** (negative feedback).
- **Aldosterone** is regulated mainly by the **renin–angiotensin–aldosterone system (RAAS)** and **potassium level**.

Cushing Syndrome

Syndrome vs. disease (instructor emphasis: know the distinction): **Cushing syndrome** = the **broad term** for **excess glucocorticoid effects over time**, from **any** cause. **Cushing's disease** = one **specific cause**: an **ACTH-producing pituitary adenoma** that keeps stimulating the adrenals.

Etiology	• **Long-term glucocorticoid medications**: the drug causes the excess. The body may **reduce its own ACTH and cortisol** in response. • **ACTH-producing pituitary adenoma** (= Cushing's disease) • **Adrenal tumor** making cortisol on its own: the pituitary **lowers ACTH**, but the tumor keeps producing • **Tumors elsewhere** in the body that release ACTH The common problem: **prolonged glucocorticoid excess**, regardless of source.
Pathophysiology and manifestations	Cortisol's normal job of making energy available during stress becomes harmful when excessive: • Liver makes **more glucose** + tissues become **insulin resistant** → **hyperglycemia, diabetes** • **Protein breakdown** → **muscle weakness**, especially **hips and shoulders** (trouble climbing stairs or rising from a chair); **thin arms and legs** • Skin loses supportive tissue → **thin skin**, **easy bruising**, **poor wound healing**, **wide purple stretch marks (striae)** • **Fat redistribution** to the **abdomen, face, and upper back** • **Weak bones** (fractures) and **suppressed immunity** (infection) • Changes in **sleep and mood**
Diagnosis	Tests check whether **normal regulation has been lost**: **late-night salivary cortisol**, **24-hour urinary free cortisol**, **dexamethasone suppression test**. You don't need the logistics. Understand that **cortisol normally falls at night** and **drops in response to a suppressing signal**; in Cushing's it doesn't. Once excess is confirmed, **ACTH identifies the source**: • **Low ACTH** → the **adrenal gland is producing cortisol on its own** • **High or "inappropriately normal" ACTH** → an **ACTH-driven** process (pituitary or another tumor)
Impact on health	Long-term: **hypertension**, **diabetes**, **osteoporosis**, **infection risk**.

Pause and Reflect: Why do clients with Cushing syndrome often develop type 2 diabetes?

Excess cortisol raises glucose **two ways**:

1. **More glucose production by the liver**. Protein breakdown also supplies amino acids the liver turns into glucose.
2. **Insulin resistance**: tissues respond less to insulin, so they take up less glucose.

More glucose enters the blood while tissues have trouble taking it up. The pancreas **compensates at first** with more insulin; if it can't keep up → **persistent hyperglycemia and diabetes**. When cortisol excess causes the diabetes, it is called **secondary diabetes** (the primary cause of type 2 is insulin resistance itself).

Addison's Disease (Primary Adrenal Insufficiency)

Etiology	The **adrenal cortex is damaged** and can't make enough of its hormones. **Autoimmune** destruction is an important cause CLARIFIED.
Pathophysiology and manifestations	↓ **Cortisol** → persistent **fatigue and weakness**, **loss of appetite**, **weight loss**, **nausea**, abdominal discomfort; impaired ability to **maintain BP** and **respond to illness**. ↓ **Aldosterone** (often) → kidneys **lose sodium** (water follows) and **retain potassium** → **low circulating volume** → **hypotension**, **dizziness on standing**, **hyponatremia**, **hyperkalemia**. **Darker skin (hyperpigmentation)**: low cortisol → pituitary raises **ACTH** → high ACTH stimulates **pigment-producing receptors**.
Diagnosis	Measure **cortisol and ACTH**. **ACTH stimulation test**: give synthetic ACTH and check whether cortisol rises. A damaged cortex **can't respond**. (Know the lab relationships, not the test logistics.)
Impact on health	**Adrenal crisis**: not enough cortisol, often during **illness or physical stress** → **severely low BP**, plus volume depletion → **shock**.

Addison's vs. a pituitary (central) cause of adrenal insufficiency:

Addison's (primary: adrenal cortex damaged)	Pituitary cause (not enough ACTH)
↓ Cortisol **and often** ↓ **aldosterone**	↓ Cortisol, but **aldosterone generally preserved** (it's controlled by RAAS and K^+, not ACTH)
↑ **ACTH** → **hyperpigmentation**	↓ **ACTH** → **no** ACTH-related hyperpigmentation
Hyponatremia, **hyperkalemia**, hypovolemia	Called "adrenal insufficiency," but **not** Addison's disease

Pause and Reflect: Why does uncontrolled Addison's lead to increased ACTH and MSH?

Use the **negative feedback loop**. Normally, circulating cortisol **holds back** the hypothalamus and pituitary. In Addison's, cortisol **stays low**, so that brake is released: the hypothalamus increases its signal and the pituitary **increases ACTH**, trying to stimulate a damaged cortex that **can't respond**. Cortisol stays low while ACTH rises.

ACTH and **MSH** (melanocyte-stimulating hormone) come from the **same large precursor molecule** (POMC, not tested). Making more precursor makes more ACTH **and** pigment-stimulating peptides, and high ACTH itself also activates pigment receptors → **darker skin**, especially in **skin creases, scars, and the oral mucosa**.

Hyperaldosteronism

The adrenal glands make **too much aldosterone**. Remember its kidney role: **hold sodium, excrete potassium**.

Pathophysiology and manifestations	• Kidneys retain **more sodium** → **water follows** → **expanded volume** → **hypertension** (the **main finding** to remember) • Kidneys lose **more potassium** → **hypokalemia** → **muscle weakness, cramps, fatigue**. An important clue, but **not present in everyone**. • **Hydrogen ion loss** → **metabolic alkalosis** • **Serum sodium is often normal**: total body sodium is higher, but the extra water dilutes it. The extra total sodium still explains the volume expansion and hypertension.
Etiology	• **Primary hyperaldosteronism**: the adrenals make aldosterone **independently** of normal signals, from a **benign aldosterone-producing tumor** or **overactive tissue in both glands**. • **Renin-driven (secondary)**: the adrenals respond to **increased renin** from the kidneys. Example: a **narrowed renal artery** → less kidney blood flow → RAAS activated → aldosterone rises CLARIFIED .
Diagnosis	Look at **aldosterone and renin together**: • **Primary:** volume expansion should **suppress renin**, so **aldosterone is inappropriately high despite low renin** → **elevated aldosterone-to-renin ratio** • **Renin-driven: both renin and aldosterone high** Medications, potassium level, and testing conditions affect results. Focus on the **relationship**, not numerical cutoffs. More testing determines whether one or both adrenals are involved.
Impact on health	Ongoing **cardiovascular and kidney damage** from increased volume and **hypertension** CORRECTED ; ↑ risk of **stroke, atrial fibrillation, heart failure**. **Low potassium** affects muscle function and **cardiac electrical activity**.
Pharm connection	**Spironolactone** and **eplerenone block the aldosterone receptor**, reducing these effects.

Quick Review

Disorder	Core problem	Hallmark findings	Key labs / complications
Pituitary adenoma	Benign anterior pituitary tumor: hormone excess and/or mass effect	Headache, peripheral (outer) vision loss, other hormone deficiencies	MRI, visual fields; apoplexy (rare, sudden)
Hyperprolactinemia	Loss of dopamine restraint, or prolactinoma	Absent/irregular periods, ↓libido, ED, infertility, galactorrhea	↑Prolactin; bone loss
Hypopituitarism	Too little of ≥1 pituitary hormone	Depends on which: hypothyroid, low cortisol, reproductive, ↓muscle, dilute urine	Cortisol deficiency is life-threatening during stress
Cushing syndrome	Prolonged glucocorticoid excess (drugs, pituitary, adrenal, other tumor)	Hyperglycemia, proximal weakness, thin skin, purple striae, central fat, thin limbs	Cortisol fails to fall at night or suppress; ↓ACTH = adrenal, ↑ACTH = ACTH-driven
Addison's	Adrenal cortex destroyed (often autoimmune): ↓cortisol, ↓aldosterone	Fatigue, weight loss, hypotension, orthostatic dizziness, darker skin	↓Na^+, ↑K^+, ↑ACTH, no response to ACTH stim; adrenal crisis
Hyperaldosteronism	Too much aldosterone: holds Na^+/water, dumps K^+/H^+	**Hypertension**; weakness or cramps if K^+ is low	Primary: ↑aldo, ↓renin. Renin-driven: both ↑. Hypokalemia, alkalosis

Self-Check ADDED

1. Where is ADH made, and where is it released?
   Made in the hypothalamus; stored and released by the posterior pituitary.

2. Why does a pituitary tumor cause loss of the outer (peripheral) visual fields?
   It presses upward on the optic chiasm, where fibers carrying peripheral vision from both eyes cross.
3. How can a client have both excess of one hormone and deficiency of others from one adenoma?
   A functioning adenoma makes excess of one hormone while compressing healthy pituitary tissue that makes the others.
4. Why can some medications raise prolactin?
   Dopamine normally inhibits prolactin. Drugs that block dopamine remove that restraint.
5. Why does high prolactin cause absent periods or infertility?
   It suppresses GnRH → ↓FSH/LH → ↓sex hormones.
6. A client's hypopituitarism suddenly causes large volumes of dilute urine. Which hormone is missing?
   ADH (hypothalamus or posterior pituitary is involved).
7. Cushing syndrome vs. Cushing's disease?
   Syndrome: glucocorticoid excess from any cause. Disease: specifically an ACTH-producing pituitary adenoma.
8. Cortisol is high and ACTH is low. Where is the problem?
   The adrenal gland is making cortisol on its own (e.g., adrenal tumor). Long-term glucocorticoid medication also lowers ACTH.
9. Why do Cushing's clients have weak hips and shoulders?
   Cortisol breaks down muscle protein.
10. Why does Addison's cause hyperkalemia and hyponatremia, but a pituitary cause of adrenal insufficiency usually doesn't?
    In Addison's, aldosterone is lost too. In a pituitary cause, aldosterone is preserved because RAAS and K^+ control it, not ACTH.
11. Why is hyperpigmentation seen in Addison's but not with a pituitary cause?
    Addison's has high ACTH (and MSH) from lost feedback. A pituitary cause has low ACTH.
12. In primary hyperaldosteronism, why is renin low?
    The expanded volume from sodium and water retention suppresses renin release.
13. Why might serum sodium be normal in hyperaldosteronism?
    Water follows the retained sodium and dilutes it, even though total body sodium is higher.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Pituitary and Adrenal Glands, NURA 340) and the lecture transcript. The slides contain only headings and images, so content comes from the narration.

- CORRECTED **"Hypotension" in hyperaldosteronism.** The transcript says health concerns come from "increased blood volume and hypotension." Excess aldosterone causes **hypertension**, which the lecturer names as the main finding earlier in the same section. This is a speech or captioning slip.
- CLARIFIED **"A type of tumor, but not a neoplasm."** The lecturer describes a pituitary adenoma this way. Technically an adenoma *is* a neoplasm (a new growth), but a **benign** one. The point being made is that it is **not cancer**, which is how the guide presents it.
- CLARIFIED **Renin-driven hyperaldosteronism.** The transcript says "Over time, the adrenal glands are responding to increased renin." In context, the lecturer is describing a **second cause** ("other times"), usually called **secondary hyperaldosteronism**, with renal artery narrowing as the example. "R**S" in the captions is RAAS.
- CLARIFIED **Autoimmune Addison's.** The transcript says Addison's is "an important cause in autoimmune dysfunction." The intended meaning is that **autoimmune destruction is an important cause of Addison's** (the most common cause in developed countries).
- CLARIFIED **Spelling.** "Sellar tarnica" in the captions = **sella turcica**.
- ADDED **"Bitemporal hemianopia"** is the formal name for the vision loss the lecturer describes (outer sides of both visual fields). You can describe it in plain words on the exam.
- ADDED **Gland and hormone table**, **Addison's vs. pituitary-cause comparison**, and **Quick Review** reorganize lecture content; the **Self-Check questions** are my own.
- Patient photos on the Cushing's slides (round face, purple abdominal striae) are not reproduced; the findings are described in the text.